

Konstantin Gerbig

Schmidt AI in Science Fellow | Department for Astronomy & Astrophysics, University of Chicago
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Employment

University of Chicago, Department for Astronomy & Astrophysics 2025–
Schmidt AI in Science Fellow, 2025 –

Yale University, Department of Astronomy 2020 – 2025
Graduate Researcher & PhD Candidate, 2020 – 2025
Teaching Assistant, 2021 – 2025

Max Planck Institute for Astronomy 2016 – 2020
Research Assistant, 2019 – 2020
Student Researcher, 2016 – 2018

UC Santa Cruz, Department for Astronomy & Astrophysics 2018 – 2019
Junior Scientist

Education

PhD, Astronomy, Yale University 2025
Advisor: Greg Laughlin. Dissertation: *"A Multiscale Perspective on Planet Formation"*
M.Sc., Astronomy, Yale University 2023
MPhil, Astronomy, Yale University 2023
M.Sc., Physics, Ruprecht-Karls Universität Heidelberg 2020
Advisors: Hubert Klahr, Ruth Murray-Clay. Thesis: *"Planetesimal Formation Regulated by Scales of Streaming and Kelvin-Helmholtz Instability"*
B.S., Physics, Ruprecht-Karls Universität Heidelberg 2017

Publications

188 citations on ADS (126 lead-author) as of type-setting this CV; h-index = 6

Lead Author:

7. **Gerbig, K.**, & Lin, M., 2026, *Diffusive Instabilities in Dusty Disks: Linear Growth and Nonlinear Breakdown*, The Astrophysical Journal, **1000**, 135
6. **Gerbig, K.**, Rice, M., Zanazzi, J.J., Christian, S., & Vanderburg, A., 2024, *Aligning Planet-Hosting Binaries via Dissipative Precession in Circumstellar Disks*, The Astrophysical Journal, **972**, 161
5. **Gerbig, K.**, Lin, M., & Lehmann, M., 2024, *Filament Formation due to Diffusive Instabilities in Dusty Protoplanetary disks*, The Astrophysical Journal, **961**, 183
4. **Gerbig, K.**, & Li, R., 2023, *Planetesimal Initial Mass Functions following Diffusion regulated Gravitational Collapse*, The Astrophysical Journal, **949**, 81
3. **Gerbig, K.**, & Laughlin, G., 2022, *The Prospects for Hurricane-like Vortices in Protoplanetary Disks*, The Astrophysical Journal, **930**, 68
2. **Gerbig, K.**, Murray-Clay, R. A., Klahr, H. & Baehr, H., 2020, *Requirements for Gravitational Collapse in Planetesimal Formation — The Impact of Scales set by Kelvin-Helmholtz and Nonlinear Streaming Instability*, The Astrophysical Journal, **895**, 91
1. **Gerbig, K.**, Lenz, C. T., & Klahr, H., 2019, *Linking Planetesimal and Dust Content in Protoplanetary*

As Contributing Author:

6. Trierweiler I., **Gerbig, K.**, & Rice M., 2025, *Constraining the Occurrence of ZLK-Induced White Dwarf Pollution with Dissipative Precession*, The Astrophysical Journal, **997**, 169
5. Cui, C., **Gerbig, K.**, Li, Y., Xu, Z., Li, R. Yu, C., Lin, M., & Yuan, F., 2025, *Dust Growth in ALMA Rings. II. Dusty Rossby Wave Instability*, The Astrophysical Journal, **986**, 86
4. Hand, J., **Gerbig, K.**, & Rice, M., 2025, *The Case for Edge-On Binaries: An Avenue Toward Comparative Exoplanet Demographics*, The Astrophysical Journal Letters, **985**, 1
3. Rice, M., **Gerbig, K.**, & Vanderburg, A., 2024, *The Orbital Geometries and Stellar Obliquities of Exoplanet-hosting Multistar Systems*, The Astronomical Journal, **167**, 126
2. Rice, M., Wang, S., **Gerbig, K.**, Wang, X., Dai, F., Tyler, D., Isaacson H., & Howard A., 2023, *The Orbital Architecture of Qatar-6: A Fully Aligned 3-Body System?*, The Astronomical Journal, **165**, 65
1. Klahr, H., Delbo, M., & **Gerbig, K.**, 2022, *Constraining the Formation of MBAs: Timing of Formation and Initial Size-Frequency Distribution*, Simone Marchi. Vesta and Ceres: Insights into the Dawn of the Solar System, **Chapter 13**, p. 199

Non-Refereed:

2. Levine, W.G., **Gerbig, K.**, Loudon, E., Lu, T., Hsieh, C., O'Connor, C., Li, R., & Dong, J., 2024, *Emerging Researchers in Exoplanetary Science (ERES): Lessons Learned in Conference Organization for Early-Career Researchers*, Bulletin of the AAS **56(1)**
1. Asali, Y., **Gerbig, K.**, Ghosh, A., Lindsay, C., Shen, Z., & Geha, M., 2022, *A Standardized Framework for Collecting Graduate Student Input in Faculty Searches*, Bulletin of the AAS **54(1)**, (equal first-authorship)

Teaching

As Lecturer:

<i>Dynamics</i> , Reutlingen University, Germany	Spring 2020
<i>Material Sciences</i> , Reutlingen University, Germany	Spring 2020
<i>Strength of Materials</i> , Reutlingen University, Germany	Spring 2020

As Teaching Assistant:

ASTR 375/575: <i>Exoplanets</i> , Yale University	Spring 2025
ASTR 160: <i>Frontiers & Controversies in Astrophysics</i> , Yale University	Fall 2022
ASTR 105: <i>The Earth in its Cosmic Context</i> , Yale University	Spring 2021
ASTR 110: <i>Planets & Stars</i> , Yale University	Fall 2021
ASTR 160: <i>Frontiers & Controversies in Astrophysics</i> , Yale University	Spring 2021
AY9: <i>Introduction to Research in Astronomy & Astrophysics</i> , UC Santa Cruz	Spring 2019
<i>Physics for Non-Physicists</i> , Heidelberg University, Germany	Spring 2018
<i>Introductory Physics Lab</i> , Heidelberg University, Germany	Fall 2017
<i>Introductory Mathematics</i> , Heidelberg University, Germany	Spring 2016

Advising & Mentoring

* Primary Research Adviser

Research Mentorship

*Po-Han (Hank) Chen, National Taiwan University (undergraduate)	2026–
*Adam Cubas, University of Chicago (undergraduate)	2026–

*Madeline Maldonado Gutierrez, Barnard College (undergraduate)	2025
Joseph Hand, University of Kansas (undergraduate); published in Hand et al., 2025 .	2024–2025
Yurou (Nina) Liu, Yale University (undergraduate)	2023–2025
*Jeremiah Reynoso, Morehouse College SURF Scholar (undergraduate)	2023–2024
*Aaron Householder, Yale University (undergraduate)	2022–2023
Yajie Liang, Heidelberg University / MPIA (B.Sc. student)	2020
Marco Vetter, Heidelberg University / MPIA (B.Sc. student)	2019–2020

Professional Mentoring

Yale Astronomy Siblings Program ("Big Sib")	2020–2025
<i>Mentored undergraduate students Andy Nilipour, Hanna Adamski, and Audrey Cesene.</i>	

Service

Professional Service

Reviewer, <i>The Astrophysical Journal</i> and <i>The Astrophysical Journal Letters</i>	2023–
Schmidt AI+Science Unseminar and Summer Schools Committees	2025–
Yale Astronomy Colloquium Committee	2024–2025
Canadian Time Allocation Committee (CanTAC), Gemini Observatory	2024
Yale Astronomy Time Allocation Committee	2024
Astronomy Climate & Diversity Committee, Yale Astronomy	2020–2021

Leadership & Organization

Lead Organizer, Schmidt AI+Science Summer School	2026
Organizer, Yale Exoplanets & Stars Seminar	2022–2025
Lead Organizer, Emerging Researchers in Exoplanet Science (ERES)	2022–2023
<i>Chaired scientific organizing committee; developed double-blind abstract review process; coordinated 100+ oral and poster presentations; co-authored conference code of conduct and outreach materials. Conference supported by a \$140,000 Heising-Simons grant.</i>	
Graduate Student Assembly Representative, Yale University	2022–2023
Astronomy Student Council, Yale University	2021–2023
<i>Developed and implemented a framework for graduate student input in faculty searches, resulting in a <i>Bulletin of the AAS</i> publication, BAAS 2022, 54(1).</i>	
Astrobites Collaboration	2022–2024
Co-Author, Yale Astronomy Newsletter	2022
Leadership Team, Yale Salsa Society	2023–2025

Professional Memberships

American Astronomical Society	2020 –
Deutsche Physikalische Gesellschaft	2016 –

Presentations

†virtual

Invited Talks

Other Worlds Laboratory, Santa Cruz, CA	Jul 2024
Academia Sinica Colloquium, Taipei, Taiwan	Feb 2024
MIT Monday Afternoon Seminar, Cambridge, MA	Dec 2023
American Museum of Natural History Astro Seminar, New York, NY	Dec 2023
†Princeton University Exoplanet Seminar	Dec 2023

Institute for Advanced Study Coffee Talk, Princeton, NJ	Dec 2023
UC Santa Cruz Planetary Seminar, Santa Cruz, CA	Nov 2023
Caltech Seminar, Pasadena, CA	Nov 2023
UCLA Astronomy Lunch Seminar, Los Angeles, CA	Nov 2023
UC Berkeley TAC Seminar, Berkeley, CA	Oct 2023
Harvard CfA ITC Luncheon, Cambridge, MA	Oct 2023
University of Chicago Astrophysics Seminar, Chicago, IL	Sep 2023
Indiana University Astrophysics Tea Talk, Bloomington, IN	Sep 2023
Cornell Astrophysics Lunch, Ithaca, NY	Sep 2023
Academia Sinica Seminar, Taipei, Taiwan	Apr 2023
†University of Michigan Stars, Planets, and Formation Journal Club	Mar 2023
†University of Arizona Star & Planet Formation Group	Feb 2023
†Caltech Yuk Lunch Seminar, Pasadena, CA	Nov 2022
Universitäts-Sternwarte München Planet Formation Group, Munich, Germany	Aug 2022
Other Worlds Laboratory, Santa Cruz, CA	Jul 2022
†Planetesimal Formation Workshop (Virtual)	Nov 2020
†Building Blocks of Planets Workshop (Virtual)	Apr 2020
Pebbles, Planetesimals and Protoplanets Workshop, Ringberg, Germany	Mar 2020
UC Santa Cruz Astrophysics Seminar, Santa Cruz, CA	Jan 2020

Contributed Talks

Dynamical Synthesis of Planetary Systems, New Haven, CT	Jan 2026
Boston Area Planetary Science Meeting, Cambridge, MA	May 2025
2nd New York Area Exoplanet Meeting, New York, NY	May 2025
†Pebbles in Planet Formation, Tokyo, Japan	Jan 2025
245th American Astronomical Society Meeting, National Harbor, MD	Jan 2025
46th Bay Area Exoplanet Meeting, Santa Cruz, CA	Jul 2024
Northeast Star and Planet Formation Meeting, Westford, MA	Jul 2024
1st New York Area Exoplanet Meeting, New York, NY	Jun 2024
Northeast Star and Planet Formation Meeting, Cambridge, MA	Jun 2023
Division on Dynamical Astronomy Meeting, East Lansing, MI	May 2023
Athena++ Workshop, New York, NY	May 2023
240th American Astronomical Society Meeting, Pasadena, CA	Jun 2022
29th Bay Area Exoplanet Meeting, Santa Cruz, CA	Jun 2019

Conference Posters

Cool Stars 23, Tokyo, Japan	Jun 2026
Extreme Solar Systems V, Christchurch, New Zealand	Mar 2024
Protostars and Planets VII, Kyoto, Japan	Apr 2023
Exoplanets IV, Las Vegas, NV	May 2022
†Exoplanets III (Virtual)	Jul 2020
235th American Astronomical Society Meeting, Honolulu, HI	Jan 2020
Extreme Solar Systems IV, Reykjavik, Iceland	Aug 2019

Media Coverage

Astronomers Take a Second Look at Twin Star Systems , Yale News	2025
A New Beginning: The Search for More Temperate Tatooines , Yale News	2024
A Grad Student Seat at the Faculty Hiring Table , Astrobites	2023

What Hurricanes and Space Storms Have in Common, Yale News Q&A	2022
Humans of Yale OISS, Yale Office of International Students and Scholars	2022
The Calm Before Planet Formation?, AAS Nova	2022
The Calm Before Planet Formation?, Astrobites	2022

Outreach

Volunteer, Yale OpenLabs	2023
Speaker, Astronomy on Tap New Haven	2022
Author & Editor, Astrobites	2022–2024
Sednoids: Echoes of a Rogue Planet in the Early Solar System?	
The First Directly Imaged Binary System with Substellar Siblings	
The Kozai-Lidov Tango: The Ups and Downs of being a Polar Circumbinary Disk	
Using tides to peek into asteroid interiors	
Diffusion and depletion of carbon monoxide in disks	
Understanding the surprising narrowness of eccentric debris belts	
Young, cool, and on edge — an unstable protoplanetary disk	
Strategies for Forming Research Talk Questions	
(Mis)alignment Between Exoplanets and Binary Stars	
Don't (Forget To) Look Up	

References

Greg Laughlin, Professor of Astronomy, Yale University
Ruth Murray-Clay, Professor of Astronomy & Astrophysics, University of California Santa Cruz
Malena Rice, Assistant Professor of Astronomy, Yale University
Min-Kai Lin, Associate Research Fellow, Academia Sinica Institute of Astronomy and Astrophysics
Hubert Klahr, Professor, Max Planck Institute for Astronomy & Heidelberg University